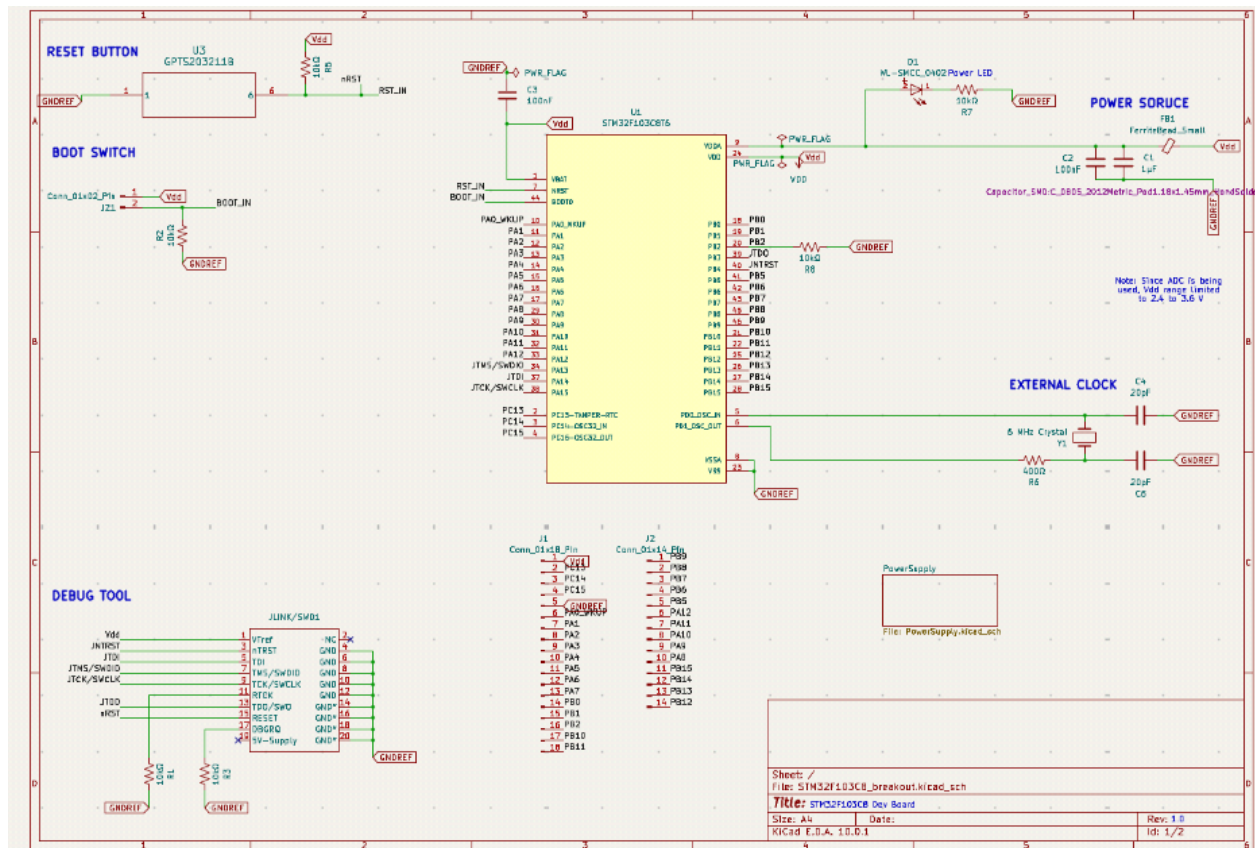


STM32F103C8T6 Dev. Board

Purpose

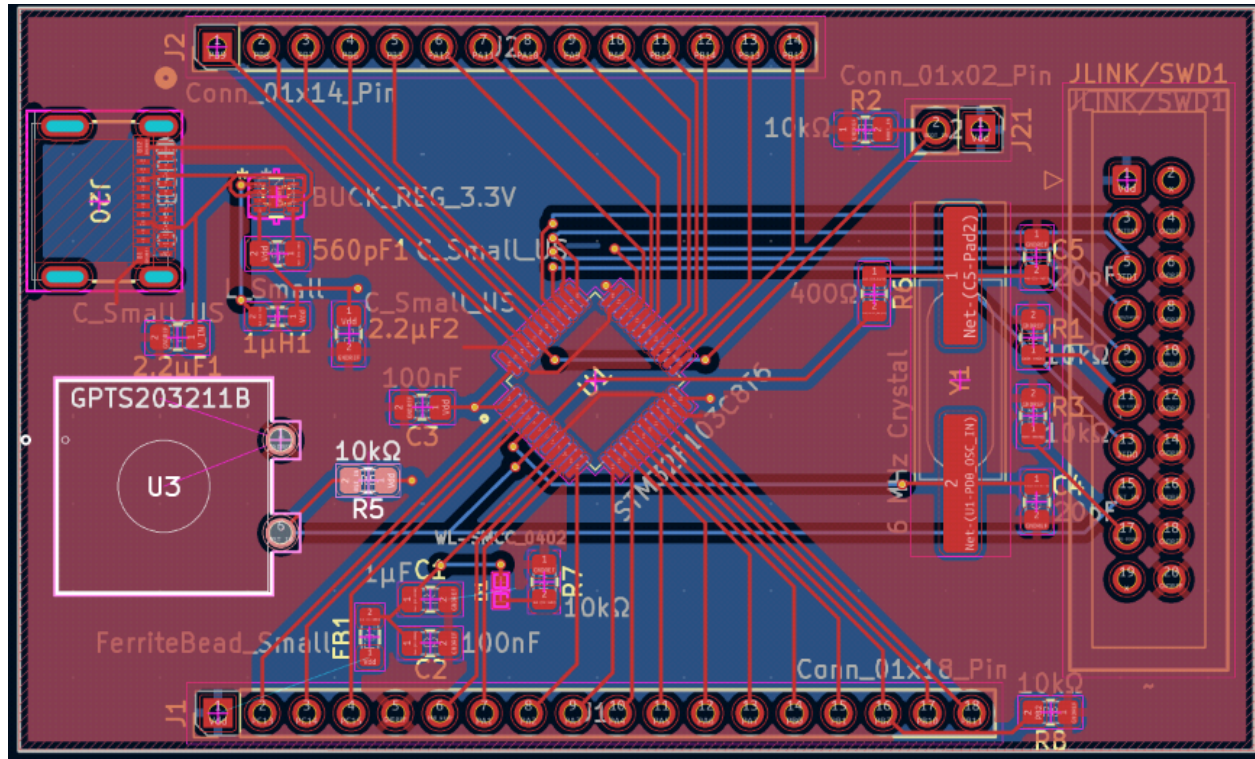
The purpose of this board is to be used as a development tool for the STM32f103C8T6 chip.

Schematic Design

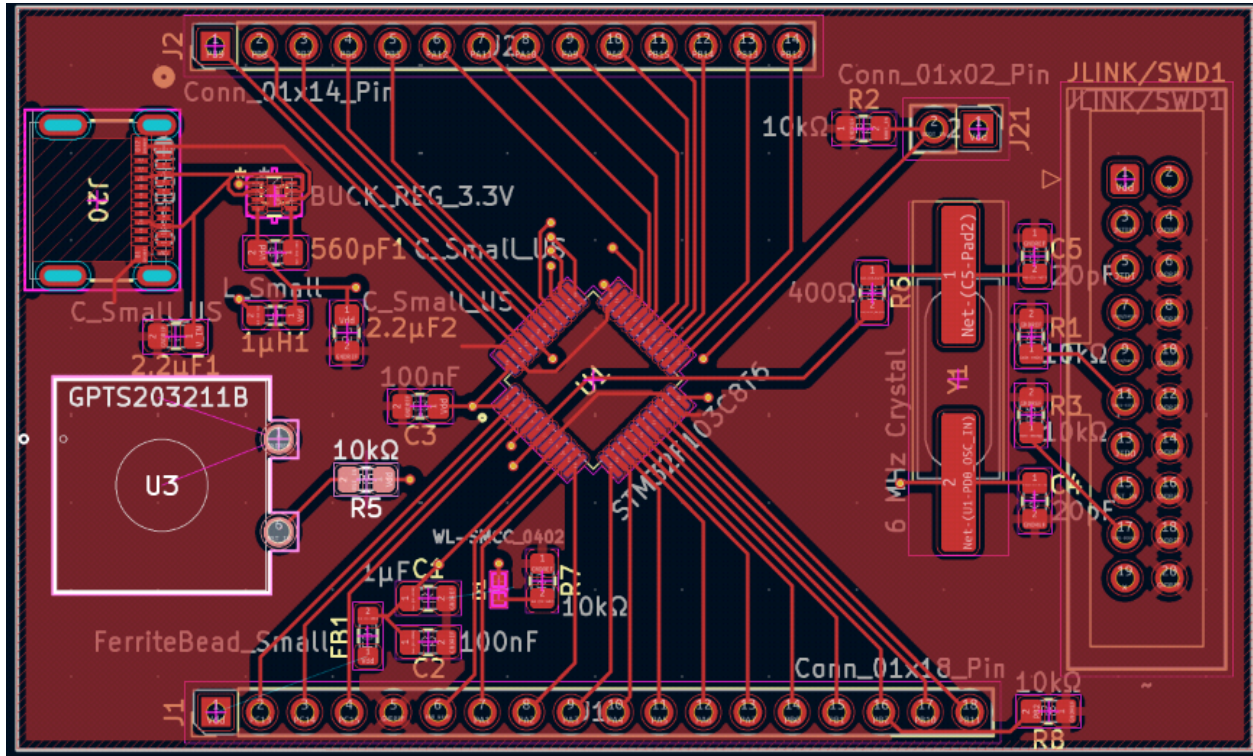


PCB Design

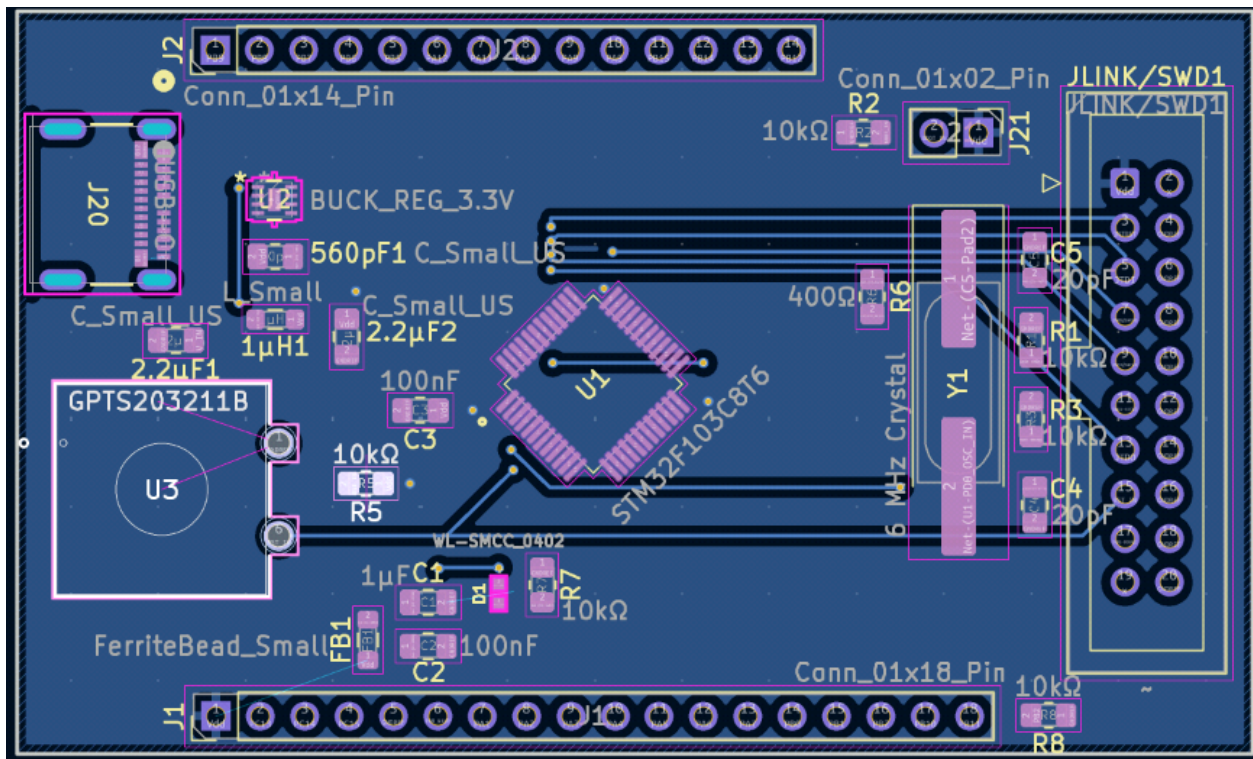
Overview



Front Layer (GNDREF)



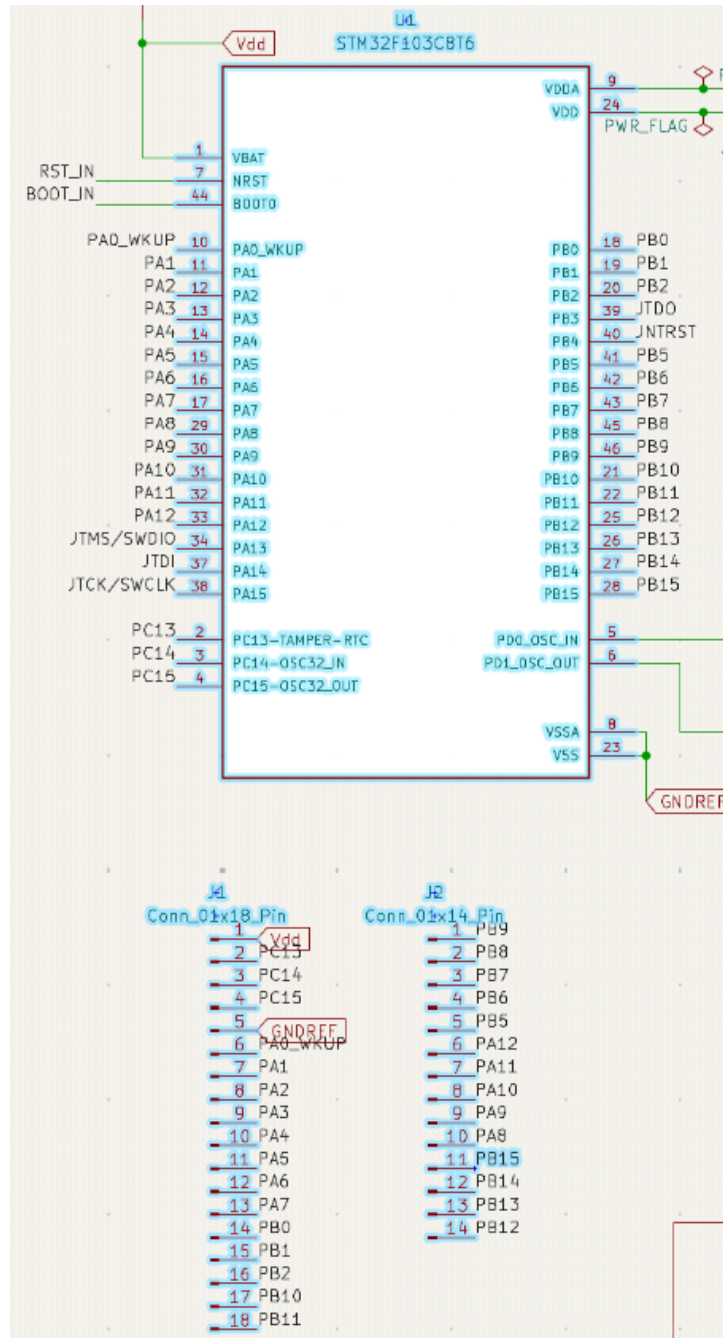
Back Layer (Vdd)



Individual Components

STM32F103C8T6 & Pin Headers

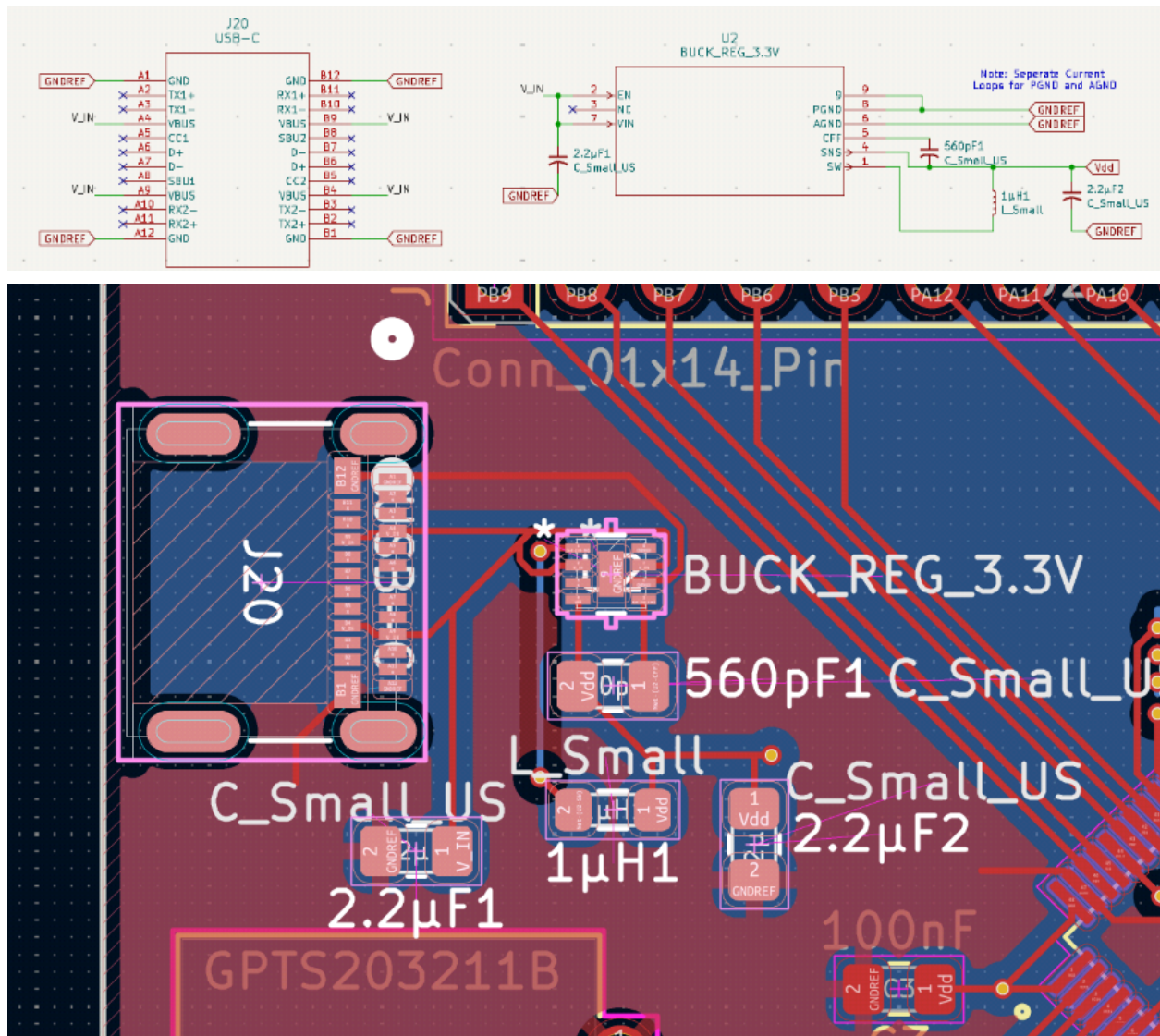
All GPIO pins that are not used internally (ex. JTAG/SWD Flash Pins) are broken out to male pin headers for general purpose use.



As you can see, the VDDA Pin is connected to the 3.3V Vdd supply. Capacitors and a Ferrite bead are used for decoupling the input for a stable supply into the STM chip, as recommended by STM [[Section 2.2 of the AN2586](#)]

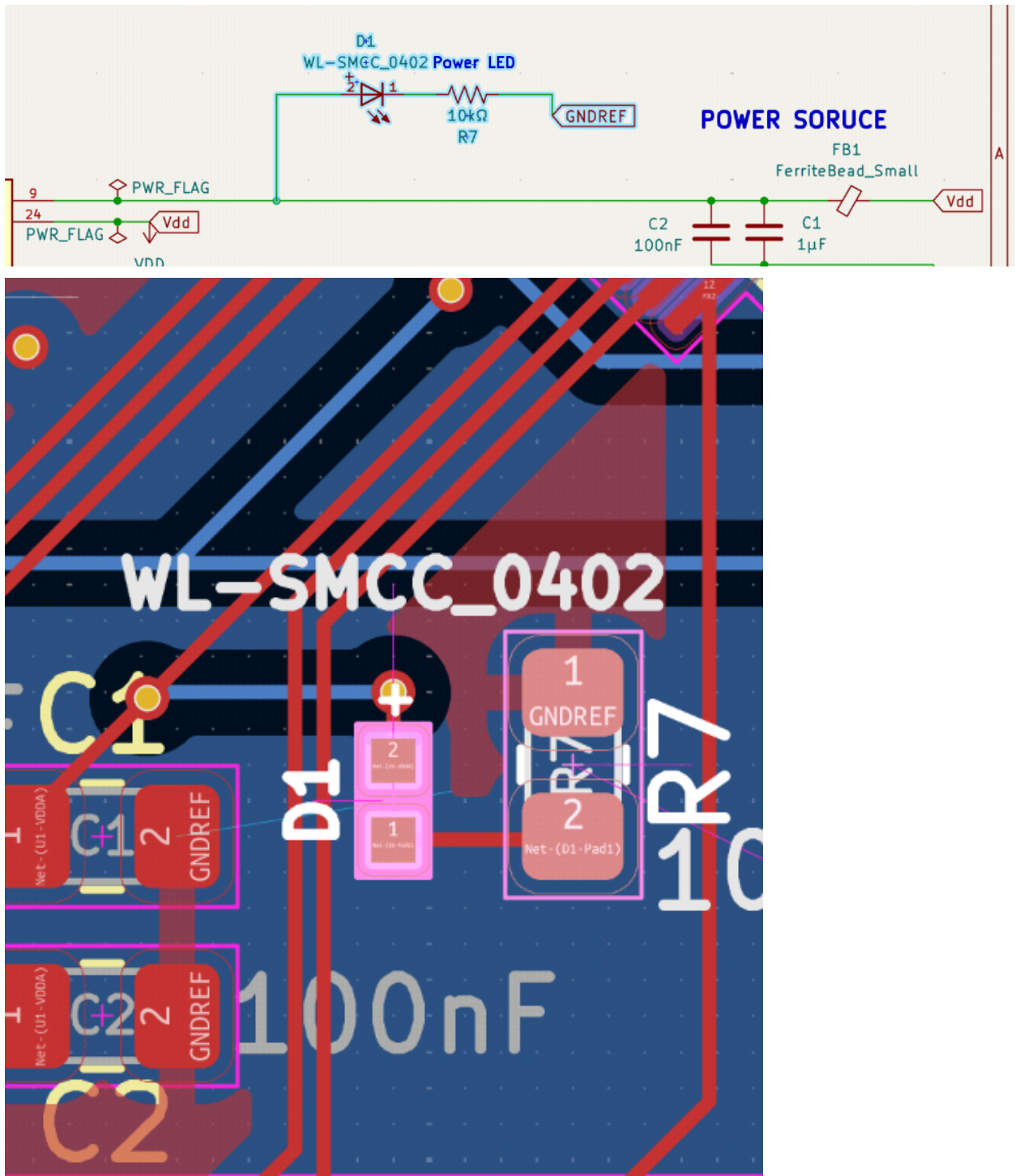
Power Supply

The development board is powered by a USB-C connection supplying a constant voltage of 5V. A Buck regulator is used to drop this down to the operating voltage of the chip (3.3V). Ground is labeled as 'GNDREF' and the power that supplies the board (output of the buck regulator) is labeled as 'Vdd'. These are global labels, allowing them to be referenced between schematic sheets, as they are here.



Power LED

The Dev. board has an LED signifying if the board is powered.

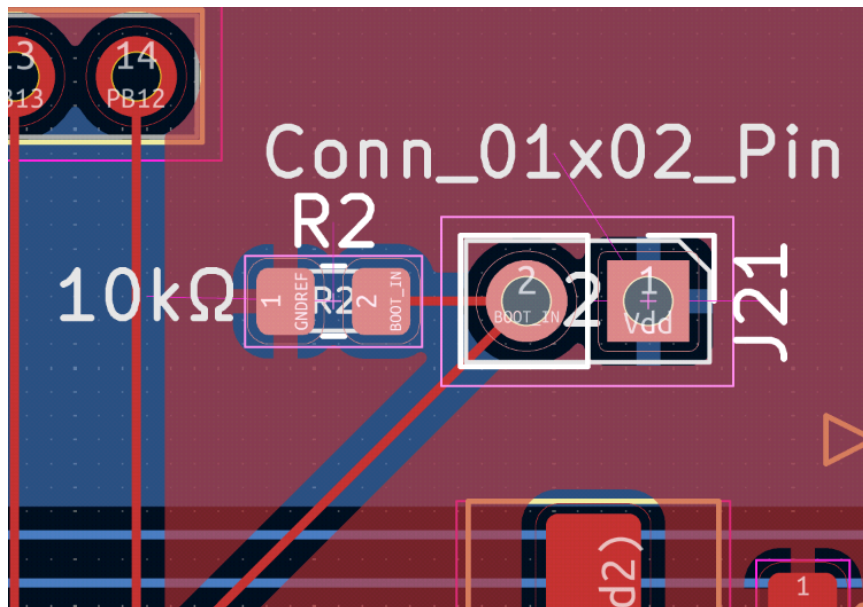
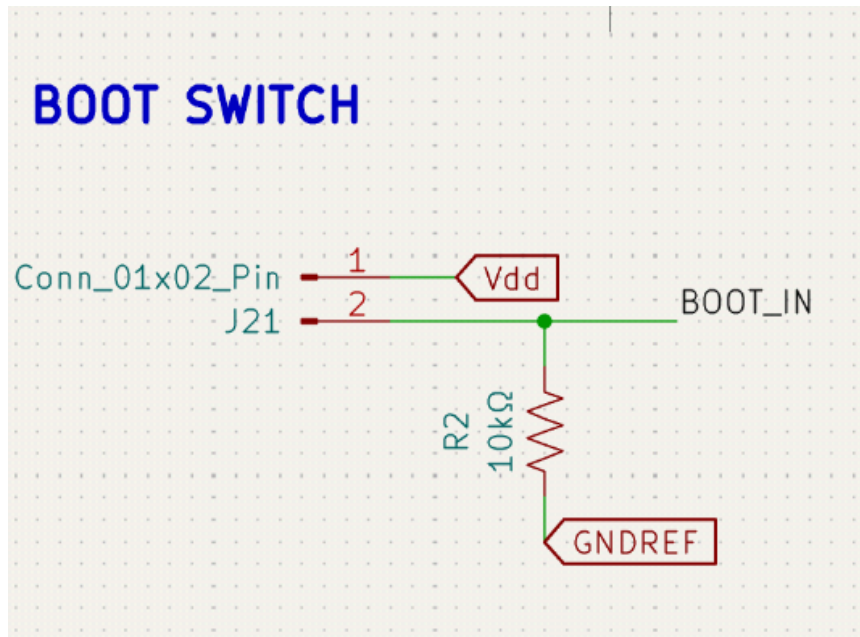


Boot Switch

This Dev. Board has 2 boot modes:

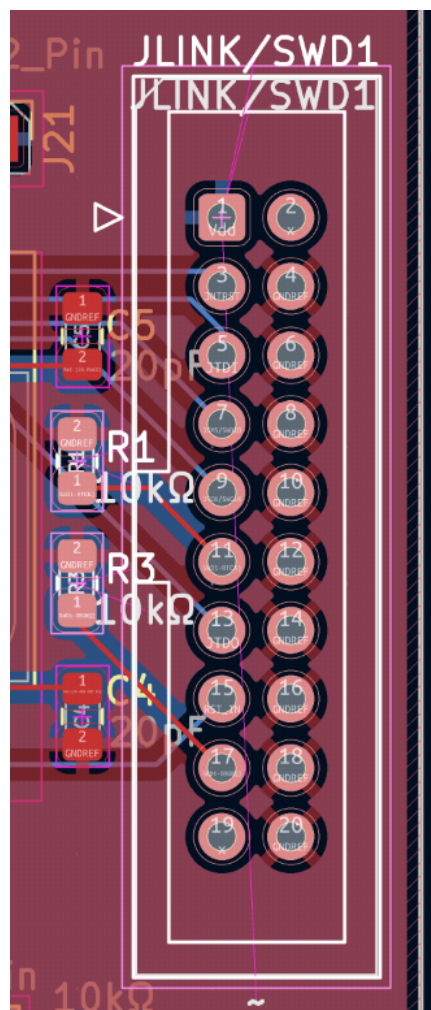
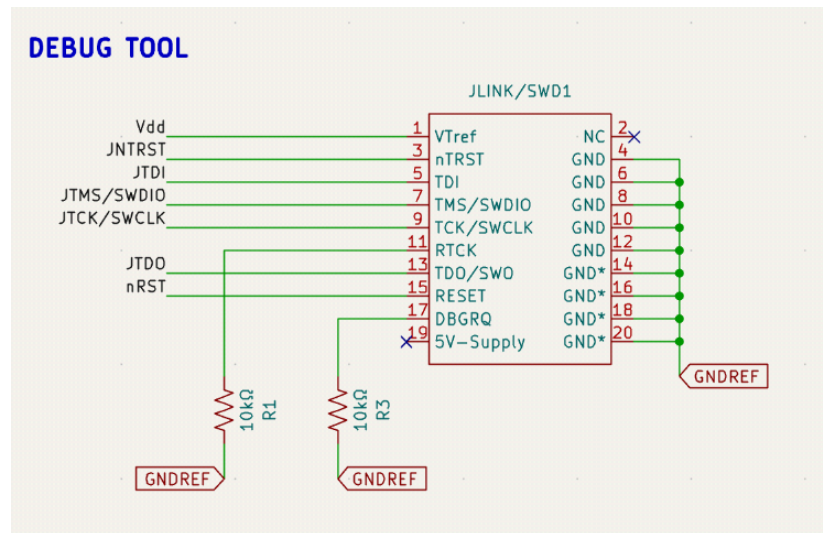
- Boot from Main Flash Memory (No Jumper, BOOT0 Pulled LOW)
- Boot from System Memory (Jumper attached, BOOT0 pulled HIGH)

Please note, BOOT0 is referred to as BOOT_IN as pin BOOT1 (also connected to PB2) is always pulled low. More information is found in [[Section 4.1 of the AN2586](#)]



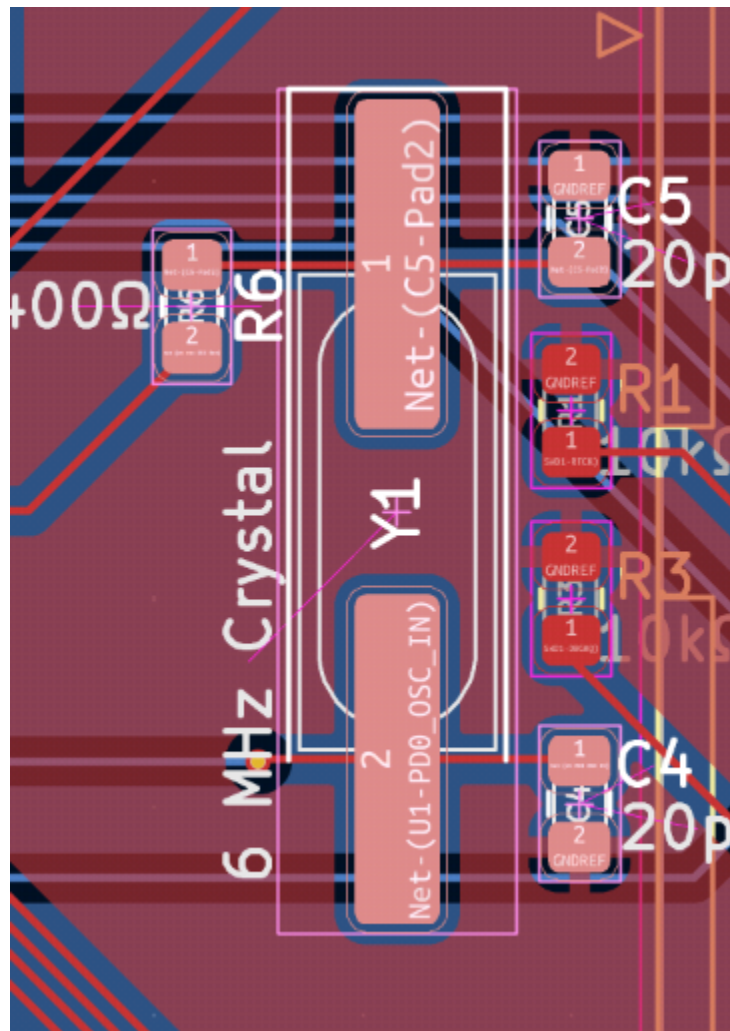
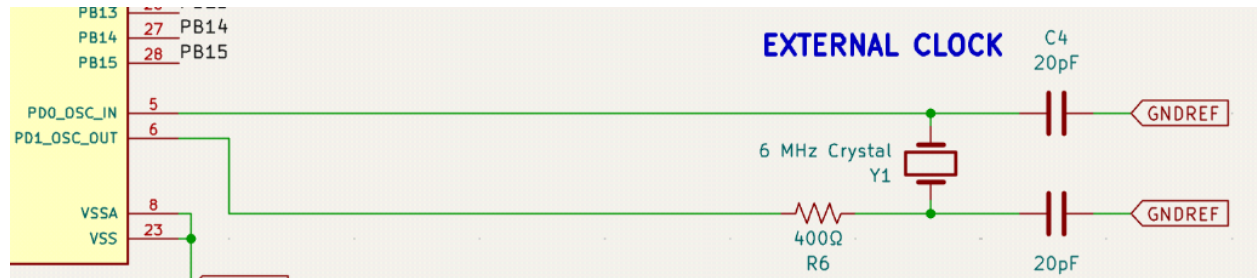
Debug Tool

The Dev. Board can be flashed & debugged using a JTAG / SWD 20-pin J-LINK Header.



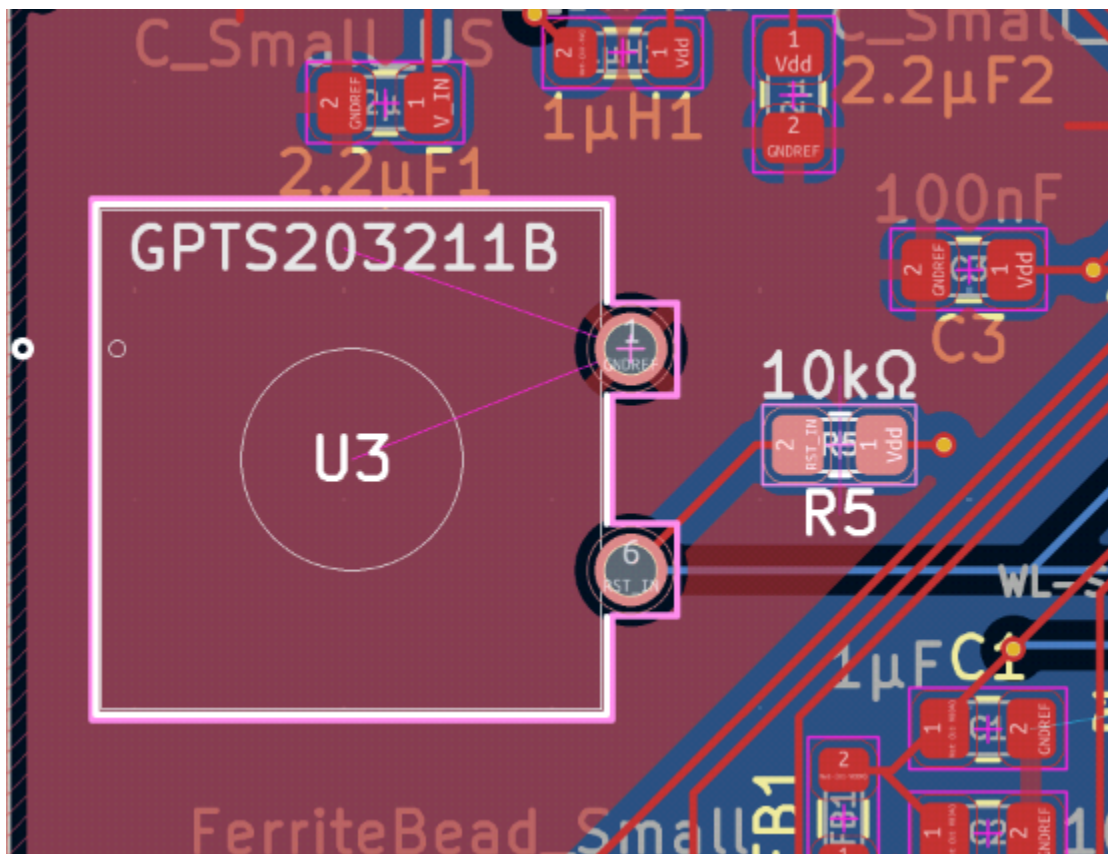
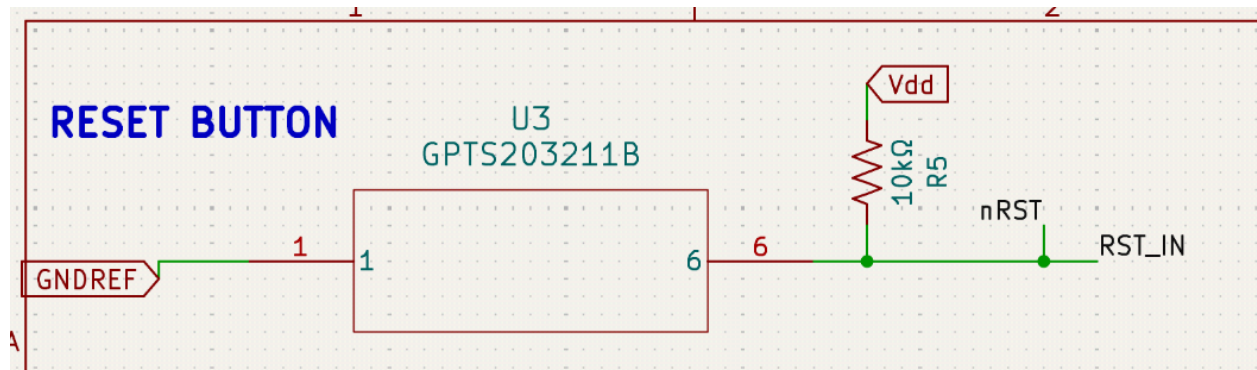
External Clock

The Dev. Board uses a dedicated external 6 MHz crystal oscillator as a clock source.



Reset Button

The Dev. Board can be reset by the press of a SPST button.



Bill of Materials

<u>Part Name</u>	<u>Part Value (Quantity)</u>	<u>Cost</u>	<u>DigiKey Link</u>	<u>Datashet</u>
Inductor	1 μ H (1)	n/a	n/a	n/a
Capacitor	2.2 μ F (2) 560 pF (1) 1 μ F (1) 100 nF (2) 20 pF (2)	n/a	n/a	n/a
Resistor	10 k Ω (6) 400 Ω (1)	n/a	n/a	n/a
Surface LED (Green)	(1)	\$0.51	https://www.digikey.com/en/products/detail/w%C3%BCrth-elektronik/150040GS73220/9857949	https://www.we-online.com/components/products/datasheet/150040GS73220.pdf
Ferrite Bead	(1)	n/a	n/a	n/a
Male Pin Headers	(34)	n/a	n/a	n/a
USB-C Female Header	(1)	\$1.61	https://www.digikey.com/en/products/detail/molex/1054500101/5843891	https://tools.molex.com/pdm_docs/PS-105448-001.pdf
J-Link 20-Pin Male Box Header	(1)	n/a	n/a	n/a
STM32F103C8T6 (Dev. Chip)	(1)	n/a	n/a	n/a
3.3V Buck Regulator	(1)	\$0.53	https://www.digikey.com/en/products/detail/microchip-technology/MIC23050-SYML-TR/1980888	https://ww1.microchip.com/downloads/en/DeviceDoc/mic23050.pdf
SPST Button	(1)	\$1.94	https://www.digikey.com/en/products/detail/cw-industries/GPTS203211B/3190590	https://www.cwind.com/sites/cwind/files/gpts203211b.pdf
6 MHz Crystal Oscillator	(1)	n/a	n/a	n/a

* (n/a) implies the components are already accessible for free.

Resources

Github Repository: <https://github.com/tymroche/KnockOut>

- Schematic & PCB In folder [STM32F103C8_breakout](#)
- Symbols & Footprints in [Symbols_Footprints](#)
- Documents & Datasheets in [Documentation](#)